

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Naveen Kumar (192521412) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1: Hospital Doctor Shift Scheduling Using Backtracking Search

Problem Statement

A hospital needs to assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) while satisfying all given constraints.

Constraints

- D1 cannot work Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor can be assigned to each shift.
- D3 cannot work Morning.
- Each doctor must be assigned exactly one shift.

Constraint Analysis

Constraint	Effect
D1 ≠ Night	D1 can only work Morning or Afternoon
D2 before D3	D2 must be assigned an earlier shift
One doctor per shift	No duplicate assignments
D3 ≠ Morning	D3 can work Afternoon or Night
One shift per doctor	Every doctor gets exactly one shift

Backtracking Search Steps

Step 1: Assign D1 = Morning.

Step 2: Assign D2 = Afternoon.

Step 3: Assign D3 = Night after verifying all constraints.

Result: Valid schedule found.

Final Assignment

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Pseudo Code

```
function Backtrack(assignments):  
  if all doctors assigned:  
    return assignments  
  choose next doctor  
  for each valid shift:  
    if constraints satisfied:
```

```
    assign shift
    if Backtrack(assignments):
        return solution
    remove assignment
return failure
```

Question 2: Robot Path Finding Using BFS

Robot moves in a 5×5 grid from Start (S) to Goal (G). Movement allowed: Up, Down, Left, Right. Cost = 1. Obstacles cannot be crossed.

Example Grid

```
S . . X .
. X . X .
. . . . .
X X . X .
. . . . G
```

BFS Steps

1. Start at S.
2. Add neighboring cells to queue.
3. Expand nodes level-by-level.
4. Ignore obstacles and visited cells.
5. Stop when Goal is reached.

Manhattan Distance

```
h(n) = |x1 - x2| + |y1 - y2|
Example: Current (2,2), Goal (4,4)
h(n)=|4-2|+|4-2|=4
```

BFS Pseudo Code

```
queue ← Start
mark Start visited
while queue not empty:
    node ← dequeue
    if node = Goal:
        return path
    for each valid neighbor:
        if not visited:
            mark visited
            enqueue neighbor
```

Question 3: Autonomous Rescue Robot Using Online Search

Constraint Analysis

- Robot moves only Up, Down, Left, Right.
- Blocked cells cannot be crossed.
- Limited sensing capability.
- Movement cost = 1, risky zones = 2.
- Environment changes dynamically.

Solution Approach

Use an Online Search strategy because the robot has incomplete knowledge and updates its map while moving.

Steps

1. Start at S.
2. Observe adjacent cells.
3. Choose the safest move.
4. Update internal map.
5. Re-plan if new obstacles appear.
6. Continue until Goal is reached.

AI Concepts

- Intelligent Agent: Senses and acts autonomously.
- Rationality: Chooses the lowest-cost safe action.
- Environment: Partially observable, dynamic, sequential, and uncertain.

Justification

Online Search is most suitable because it allows the robot to adapt to changing environments, avoid newly discovered obstacles, and safely reach the survivor.

Conclusion

Backtracking efficiently solves scheduling problems, BFS guarantees the shortest path in an unweighted grid, and Online Search is effective for dynamic real-world environments.